

Project: ProjectHub+ – Student Project Submission & Academic Portfolio Portal

A web-based portal designed to automate and manage student project submissions, certificate verification, and academic portfolio tracking in a university setting.

The system supports multiple user roles such as students, faculty, and administrators with defined permissions and structured workflows. The portal enables project uploads, faculty review and grading, certificate approval, and professional profile integration to build a verified academic portfolio.

User Authentication

(This module uses SQL – recommended for structured user credentials, passwords, and role mapping.)

- **JWT-Based Login System**

Implement token-based authentication using JWT to support stateless, secure access for all users interacting with the system via frontend or API.

- **Role-Based Access Control (RBAC)**

Define user roles with controlled access to system features:

- **Student:** Upload projects and certificates, track submission status, and manage professional profiles.
 - **Faculty:** Review submissions, verify certificates, provide feedback, and assign grades.
 - **Admin:** Full control to manage users, configure submission rules, assign faculty, and monitor analytics.
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Frontend Modules

1. Student Dashboard

- View and update personal profile.
- Upload project files (PDF / ZIP / DOC).
- Enter project details such as title, subject, abstract, and technology used.
- Save draft before submission.
- Edit/update project before approval.
- View submission history.
- Track project status (Draft / Pending Review / Approved / Rejected / Resubmitted).
- Download faculty feedback.
- Resubmit project after rejection.
- Upload certificates (PDF/Image).

- Enter certificate details (title, organization, category, issue date).
 - Track certificate verification status (Pending / Approved / Rejected).
 - Add professional profile URLs (GitHub, HackerRank, LeetCode, LinkedIn, Portfolio).
 - View academic analytics summary (total submissions, approved projects, verified certificates).
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2. Faculty Panel

- Display assigned student projects.
 - Filter projects by status or subject.
 - Search by student name or project title.
 - Download submitted project files.
 - Approve or reject project submissions.
 - Add detailed feedback and assign grades.
 - Review resubmissions with version history.
 - View student coding profile links.
 - Verify or reject uploaded certificates.
 - Add remarks for certificate validation.
 - Generate evaluation summary report.
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3. Admin Interface

- Create, edit, and delete user accounts.
 - Assign user roles and manage authentication credentials.
 - Assign faculty to subjects or student groups.
 - Configure submission deadlines and resubmission rules.
 - Monitor system statistics (total submissions, approval rate, certificate verification rate).
 - Export grade and certificate reports.
 - Maintain audit logs and activity tracking.
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4. Project Submission Module

- Auto-generated Submission ID.
 - Draft saving functionality.
 - Version tracking for resubmissions.
 - Deadline enforcement.
 - File type and size validation.
 - Status-based workflow management.
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5. Certificate Verification Module

- Auto-generated Certificate ID.
 - Upload validation for secure file storage.
 - Approval workflow (Student Upload → Faculty Review → Approval/Rejection).
 - Timestamp tracking for verification.
 - Faculty remark logging.
 - Resubmission option after rejection.
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6. Notification Module

- Real-time notifications for submission status updates.
 - Certificate approval/rejection alerts.
 - Faculty feedback notifications.
 - Deadline reminder notifications.
 - Persistent notification history.
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Backend Modules (Spring Boot + SQL + NoSQL)

(SQL is used for credential storage and structured records. NoSQL is used for flexible project and certificate data.)

1. User Authentication Service (SQL)

- REST APIs for login, logout, registration, and token management.
 - Middleware for securing endpoints based on user roles.
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2. Student APIs (NoSQL)

- Endpoints for project submission and update.
 - Fetch submission history and status tracking.
 - Certificate upload and management.
 - Profile URL management APIs.
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3. Faculty APIs (NoSQL)

- APIs to fetch assigned submissions.
- Approve/reject projects and certificates.

- Submit grades and feedback.
 - Access student profile links.
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4. Admin APIs (NoSQL)

- User management APIs.
 - Faculty assignment configuration.
 - Deadline configuration.
 - Generate academic and certificate analytics reports.
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5. Notification APIs (NoSQL)

- REST endpoints to create and fetch notifications.
 - Persistent storage for alerts.
 - Support for real-time status updates.
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Deliverables

- Secure authentication system using JWT with role-based access control.
 - Complete project submission workflow with version tracking.
 - Faculty review and grading module.
 - Certificate upload and approval system.
 - Professional profile integration module.
 - Real-time notification system.
 - Responsive frontend built with React, organized into modular components by role.
 - RESTful API backend built with Spring Boot, connected to SQL and NoSQL databases.
 - Schema definitions for users, submissions, certificates, grades, and notifications.
 - Exportable grade and certificate reports in PDF/Excel format.
 - Role-based routing and frontend route protection.
 - Backend unit testing and API documentation using Swagger/OpenAPI.
 - GitHub repository link containing the complete project source code.
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Deployment

The system is deployed using cloud services for hosting frontend, backend, and databases. The deployment process is organized into frontend deployment, backend deployment, SQL database setup, and NoSQL database setup.
